Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In this section we introduced an important difference between real and complex analysis, namely, that we cannot graph a complex function. Instead, we represent a complex function with two figures: the first a subset S in the complex plane, and the second, the image S' of the set S under a complex mapping. A complete understanding of a complex mapping is obtained when we understand the relationship between *any* set S and its image S'.

(*ii*) Complex mappings are closely related to parametric curves in the plane. In later sections, we use this relationship to help visualize the notions of limit, continuity, and differentiability of complex functions. Parametric curves will also be of central importance in the study of complex integrals much as they were in the study of real line integrals.

EXERCISES 2.2

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, proceed as in Example 1 or Example 2 to find the image S' of the set S under the given complex mapping $w = f(z)$.

1. $f(z) = i\bar{z}$; S is the horizontal line $y = 3$

2. $f(z) = \bar{z}$; S is the line $y = x$

3. $f(z) = 3z$; S is the half-plane $\text{Im}(z) > 2$

4. $f(z) = 3iz$; S is the infinite vertical strip $2 \le \text{Re}(z) < 3$

5. $f(z) = (1 + i)z$; S is the vertical line $x = 2$

6. $f(z) = (1 - i)z$; S is the line $y = 2x + 1$

7. $f(z) = iz + 4$; S is the half-plane $\text{Im}(z) \le 1$

8. $f(z) = iz + \bar{z}$; S is the vertical line $x = 3$

In Problems 9–14, find the image of the given line under the complex mapping $w = z^2$.

9. $y = 1$

10. $x = -3$

11. $x = 0$

12. $y = 0$

13. $y = x$

14. $y = -x$

In Problems 15–20, (**a**) plot the parametric curve C given by $z(t)$ and describe the curve in words, (**b**) find a parametrization of the image, C', of C under the given complex mapping $w = f(z)$, and (**c**) plot C' and describe this curve in words.

15. $z(t) = 2(1 - t) + it, 0 \le t \le 1;\ f(z) = 3z$

16. $z(t) = i(1 - t) + (1 + i)t,\ 0 \le t < \infty;\ f(z) = -z$

17. $z(t) = 1 + 2e^{it},\ 0 \le t \le 2\pi;\ f(z) = z + 1 - i$

18. $z(t) = i + e^{it},\ 0 \le t \le \pi;\ f(z) = (z - i)^2$

19. $z(t) = t,\ 0 \le t \le 2;\ f(z) = e^{i\pi z}$

20. $z(t) = 4e^{it},\ 0 \le t \le \pi,\ f(z) = \text{Re}(z)$

In Problems 21–26, use parametrizations to find the image, C', of the curve C under the given complex mapping $w = f(z)$.

21. $f(z) = z^3$; C is the positive imaginary axis

22. $f(z) = iz$; C is the circle $|z - 1| = 2$

23. $f(z) = 1/z$; C is the circle $|z| = 2$

24. $f(z) = 1/z$; C is the line segment from $1 - i$ to $2 - 2i$

25. $f(z) = z + \bar{z}$; C is the semicircle of the unit circle $|z| = 1$ in the upper half-plane $\text{Im}(z) \geq 0$

26. $f(z) = e^z$; C is the ray emanating from the origin and containing $2 + \sqrt{3}i$

Focus on Concepts

27. In this problem we will find the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**a**) The line $x = 1$ consists of all points $z = 1 + iy$ where $-\infty < y < \infty$. Find the real and imaginary parts u and v of $f(z) = 1/z$ at a point $z = 1 + iy$ on this line.

(**b**) Show that $\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}$ for the functions u and v from part (a).

(**c**) Based on part (b), describe the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**d**) Is there a point on the line $x = 1$ that maps onto 0? Do you want to alter your description of the image in part (c)?

28. Consider the parametrization $z(t) = i(1 - t) + 3t$, $0 \leq t \leq 1$.

(**a**) Describe in words this parametric curve.

(**b**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = 3(1 - t) + it$, $0 \leq t \leq 1$?

(**c**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = \frac{3}{2}t + i\left(1 - \frac{1}{2}t\right)$, $0 \leq t \leq 2$?

(**d**) Find a parametrization of the line segment from $1 + 2i$ to $2 + i$ where the parameter satisfies $0 \leq t \leq 3$.

29. Use parametrizations to find the image of the circle $|z - z_0| = R$ under the mapping $f(z) = iz - 2$. Describe the image in words.

30. Consider the line $y = mx + b$ in the complex plane.

(**a**) Give a parametrization $z(t)$ for the line.

(**b**) Describe in words the image of the line under the complex mapping $w = z + 2 - 3i$.

(**c**) Describe in words the image of the line under the complex mapping $w = 3z$.

31. The complex mapping $w = \bar{z}$ is called **reflection** about the real axis. Explain why.

32. Let $f(z) = az$ where a is a complex constant and $|a| = 1$.

(**a**) Show that $|f(z_1) - f(z_2)| = |z_1 - z_2|$ for all complex numbers z_1 and z_2.

(**b**) Give a geometric interpretation of the result in (a).

(**c**) What does your answer to (b) tell you about the image of a circle under the complex mapping $w = az$.

33. In this problem we investigate the effect of the mapping $w = az$, where a is a complex constant and $a \neq 0$, on angles between rays emanating from the origin.

(**a**) Let C be a ray in the complex plane emanating from the origin. Use parametrizations to show that the image C' of C under $w = az$ is also a ray emanating from the origin.

(**b**) Consider two rays C_1 and C_2 emanating from the origin such that C_1 contains the point $z_1 = a_1 + ib_1$ and C_2 contains the point $z_2 = a_2 + ib_2$. In multivariable calculus, you saw that the angle θ between the rays C_1 and C_2 (which is the same as the angle between the position vectors $(a_1,\ b_1)$ and $(a_2,\ b_2)$) is given by:

$$\theta = \arccos\left(\frac{a_1a_2 + b_1b_2}{\sqrt{a_1^2 + b_1^2}\sqrt{a_2^2 + b_2^2}}\right) = \arccos\left(\frac{z_1\bar{z}_2 + \bar{z}_1z_2}{2\,|z_1|\,|z_2|}\right). \tag{14}$$

Let C_1' and C_2' be the images of C_1 and C_2 under $w = az$. Use part (a) and (14) to show that the angle between C_1' and C_2' is the same as the angle between C_1 and C_2.

34. Consider the complex mapping $w = z^2$.

(**a**) Repeat Problem 33(a) for the mapping $w = z^2$.

(**b**) Experiment with different rays. What effect does the complex mapping $w = z^2$ appear to have on angles between rays emanating from the origin?

Computer Lab Assignments

In Problems 35–38, use a CAS to (**a**) plot the image of the unit circle under the given complex mapping $w = f(z)$, and (**b**) plot the image of the line segment from 1 to $1 + i$ under the given complex mapping $w = f(z)$.

35. $f(z) = z^2 + (1 + i)z - 3$

36. $f(z) = iz^3 + z - i$

37. $f(z) = z^4 - z$

38. $f(z) = z^3 - \bar{z}$

2.3 Linear Mappings

INTRODUCTION Recall that a real function of the form $f(x) = ax + b$ where a and b are any *real* constants is called a linear function. In keeping with the similarities between real and complex analysis, we define a **complex linear function** to be a function of the form $f(z) = az + b$ where a and b are any *complex* constants. Just as real linear functions are the easiest types of real functions to graph, complex linear functions are the easiest types of complex functions to visualize as mappings of the complex plane. In this section, we will show that every nonconstant complex linear mapping can be described as a composition of three basic types of motions: a translation, a rotation, and a magnification.

Before looking at a general complex linear mapping $f(z) = az + b$, we investigate three special types of linear mappings called translations, rotations, and magnifications. Throughout this section we use the symbols T, R, and M to represent mapping by translation, rotation, and magnification, respectively.

Translations

A complex linear function

$$T(z) = z + b, \quad b \neq 0, \tag{1}$$

is called a **translation**. If we set $z = x + iy$ and $b = x_0 + iy_0$ in (1), then we obtain:

$$T(z) = (x + iy) + (x_0 + iy_0) = x + x_0 + i(y + y_0).$$

Thus, the image of the point $(x,\ y)$ under T is the point $(x + x_0,\ y + y_0)$. From FIGURE 2.3.1 we see that if we plot $(x,\ y)$ and $(x + x_0,\ y + y_0)$ in the same copy of the complex plane, then the vector originating at $(x,\ y)$ and terminating at $(x + x_0,\ y + y_0)$ is $(x_0,\ y_0)$; equivalently, if we plot z and $T(z)$ in the same copy of the complex plane, then the vector originating at z and terminating at $T(z)$ is b. Therefore, the linear mapping $T(z) = z + b$ can be visualized in a single copy of the complex plane as the process of *translating* the point z along the vector b to

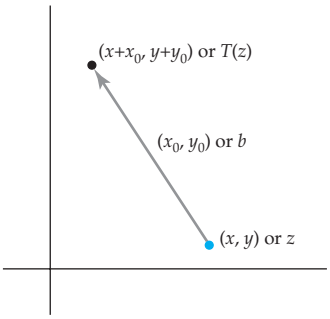


FIGURE 2.3.1 Translation